

2.1)

```
C:\Users\Siu13>ping 142.232.76.200

Pinging 142.232.76.200 with 32 bytes of data:
Reply from 142.232.76.200: bytes=32 time=4ms TTL=59
Reply from 142.232.76.200: bytes=32 time=6ms TTL=59
Reply from 142.232.76.200: bytes=32 time=6ms TTL=59
Reply from 142.232.76.200: bytes=32 time=5ms TTL=59

Ping statistics for 142.232.76.200:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 4ms, Maximum = 6ms, Average = 5ms
```

3.1)

```
C:\Users\Siu13>nslookup example.com
Server:  ns1.bcit.ca
Address:  142.232.76.200

Non-authoritative answer:
Name:     example.com
Addresses: 2600:1406:5e00:6::17ce:bc12
           2600:1406:5e00:6::17ce:bc1b
           2600:1406:bc00:53::b81e:94c8
           2600:1406:bc00:53::b81e:94ce
           2600:1408:ec00:36::1736:7f24
           2600:1408:ec00:36::1736:7f31
           23.220.75.245
           23.192.228.80
           23.192.228.84
           23.215.0.136
           23.215.0.138
           23.220.75.232
```

3.2)

```
iu13@Nathan-Laptop:/mnt/c/Users/Siu13$ dig example.com A

; <<>> DiG 9.18.39-0ubuntu0.24.04.1-Ubuntu <<>> example.com A
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 1683
;; flags: qr rd ra; QUERY: 1, ANSWER: 6, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
;; QUESTION SECTION:
;example.com.                IN      A

;; ANSWER SECTION:
example.com.                239     IN      A      23.215.0.136
example.com.                239     IN      A      23.215.0.138
example.com.                239     IN      A      23.220.75.232
example.com.                239     IN      A      23.220.75.245
example.com.                239     IN      A      23.192.228.80
example.com.                239     IN      A      23.192.228.84

;; Query time: 47 msec
;; SERVER: 10.255.255.254#53(10.255.255.254) (UDP)
;; WHEN: Tue Oct 07 10:08:30 PDT 2025
;; MSG SIZE rcvd: 136
```

3.3)

```
iu13@Nathan-Laptop:/mnt/c/Users/Siu13$ dig +short example.com A
23.215.0.138
23.220.75.232
23.220.75.245
23.192.228.80
23.192.228.84
23.215.0.136
```

3.4)

No.	Time	Source	Destination	Protocol	Length	Info
2796	138.555101	142.232.76.200	10.65.97.239	DNS	112	Standard query response 0x0001 PTR 200.76.232.142.in-addr.arpa PTR ns1.bc.it.ca
2797	138.559848	10.65.97.239	142.232.76.200	DNS	71	Standard query 0x0002 A example.com
2798	138.568567	142.232.76.200	10.65.97.239	DNS	167	Standard query response 0x0002 A example.com A 23.192.228.84 A 23.215.0.136 A 23.215.0.138 A 23.220.75.232 A 23...
2799	138.579894	10.65.97.239	142.232.76.200	DNS	71	Standard query 0x0003 AAAA example.com
2800	138.584672	142.232.76.200	10.65.97.239	DNS	239	Standard query response 0x0003 AAAA example.com AAAA 2600:1406:bc00:53::b81e:94c8 AAAA 2600:1406:bc00:53::b81e:...
2812	138.788534	10.65.97.239	142.232.76.200	DNS	130	Standard query 0xc710 HTTPS 3-prod-dynamite-prod-09-us-signaler-pa.clients6.google.com
2814	138.788961	10.65.97.239	142.232.76.200	DNS	130	Standard query 0x4583 A 3-prod-dynamite-prod-09-us-signaler-pa.clients6.google.com
2819	138.792963	142.232.76.200	10.65.97.239	DNS	148	Standard query response 0x4583 A 3-prod-dynamite-prod-09-us-signaler-pa.clients6.google.com A 142.250.73.106
2820	138.792963	142.232.76.200	10.65.97.239	DNS	182	Standard query response 0xc710 HTTPS 3-prod-dynamite-prod-09-us-signaler-pa.clients6.google.com SOA ns1.google...
3021	148.066636	10.65.97.239	142.232.76.200	DNS	98	Standard query 0x3df1 HTTPS waa-pa.clients6.google.com
3021	148.067358	10.65.97.239	142.232.76.200	DNS	98	Standard query 0x4277 A waa-pa.clients6.google.com
3029	148.073439	142.232.76.200	10.65.97.239	DNS	150	Standard query response 0x3df1 HTTPS waa-pa.clients6.google.com SOA ns1.google.com
3031	148.076659	142.232.76.200	10.65.97.239	DNS	116	Standard query response 0x4277 A waa-pa.clients6.google.com A 142.250.217.74
3285	170.992451	10.65.97.239	142.232.76.200	DNS	87	Standard query 0x3f80 HTTPS ssl.gstatic.com
3287	170.992920	10.65.97.239	142.232.76.200	DNS	87	Standard query 0x76c1 A ssl.gstatic.com
3291	170.996841	142.232.76.200	10.65.97.239	DNS	146	Standard query response 0x3f80 HTTPS ssl.gstatic.com SOA ns1.google.com
3293	170.996841	142.232.76.200	10.65.97.239	DNS	105	Standard query response 0x76c1 A ssl.gstatic.com A 142.250.73.99
3342	177.153326	10.65.97.239	142.232.76.200	DNS	101	Standard query 0xcde3 A rs3clients.timeusage.family.microsoft.com

Frame 2798: 167 bytes on wire (1336 bits), 167 bytes captured (1336 bits) on interface \Device...
 Ethernet II, Src: JuniperNetwo_53:d6:a0 (d0:07:ca:53:d6:a0), Dst: Intel_bf:06:80 (fc:77:74:bf:06:80)
 Internet Protocol Version 4, Src: 142.232.76.200, Dst: 10.65.97.239
 User Datagram Protocol, Src Port: 53, Dst Port: 65094
 Domain Name System (response)

4.1)

```

C:\Users\Siu13>nslookup -type=AAAA example.com
Server: ns1.bc.it.ca
Address: 142.232.76.200

Non-authoritative answer:
Name: example.com
Addresses: 2600:1408:ec00:36::1736:7f24
           2600:1408:ec00:36::1736:7f31
           2600:1406:5e00:6::17ce:bc12
           2600:1406:5e00:6::17ce:bc1b
           2600:1406:bc00:53::b81e:94c8
           2600:1406:bc00:53::b81e:94ce
  
```

```

iu13@Nathan-Laptop:/mnt/c/Users/Siu13$ dig example.com AAAA

; <<>> DiG 9.18.39-0ubuntu0.24.04.1-Ubuntu <<>> example.com AAAA
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 12486
;; flags: qr rd ra; QUERY: 1, ANSWER: 6, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
;; QUESTION SECTION:
;example.com.                IN      AAAA

;; ANSWER SECTION:
example.com.                168     IN      AAAA    2600:1408:ec00:36::1736:7f31
example.com.                168     IN      AAAA    2600:1406:5e00:6::17ce:bc12
example.com.                168     IN      AAAA    2600:1406:5e00:6::17ce:bc1b
example.com.                168     IN      AAAA    2600:1406:bc00:53::b81e:94c8
example.com.                168     IN      AAAA    2600:1406:bc00:53::b81e:94ce
example.com.                168     IN      AAAA    2600:1408:ec00:36::1736:7f24

;; Query time: 56 msec
;; SERVER: 10.255.255.254#53(10.255.255.254) (UDP)
;; WHEN: Tue Oct 07 11:07:09 PDT 2025
;; MSG SIZE rcvd: 208

```

4.2)

```

C:\Users\Siu13>nslookup -type=MX ietf.org
Server:  ns1.bcit.ca
Address:  142.232.76.200

Non-authoritative answer:
ietf.org      MX preference = 0, mail exchanger = mail2.ietf.org

```

```
siu13@Nathan-Laptop:/mnt/c/Users/Siu13$ dig ietf.org MX

; <<>> DiG 9.18.39-0ubuntu0.24.04.1-Ubuntu <<>> ietf.org MX
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 50121
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
;; QUESTION SECTION:
;ietf.org.                IN      MX

;; ANSWER SECTION:
ietf.org.                60      IN      MX      0 mail2.ietf.org.

;; Query time: 28 msec
;; SERVER: 10.255.255.254#53(10.255.255.254) (UDP)
;; WHEN: Tue Oct 07 11:11:59 PDT 2025
;; MSG SIZE rcvd: 59
```

4.3)

```
C:\Users\Siu13>nslookup -type=NS example.com
Server:  node-1w7jr9n24twqzs2cg5ed4tjl3.ipv6.telus.net
Address:  2001:568:ff09:10c::67
```

Non-authoritative answer:

```
example.com      nameserver = b.iana-servers.net
example.com      nameserver = a.iana-servers.net
```

```
C:\Users\Siu13>nslookup -type=SOA example.com
Server:  node-1w7jr9n24twqzs2cg5ed4tjl3.ipv6.telus.net
Address:  2001:568:ff09:10c::67
```

Non-authoritative answer:

```
example.com
      primary name server = ns.icann.org
      responsible mail addr = noc.dns.icann.org
      serial      = 2025082258
      refresh     = 7200 (2 hours)
      retry       = 3600 (1 hour)
      expire      = 1209600 (14 days)
      default TTL = 3600 (1 hour)
```

```
C:\Users\Siu13>wsl
iu13@Nathan-Laptop: /mnt/c/Users/Siu13$ dig example.com NS

; <<>> DiG 9.18.39-0ubuntu0.24.04.1-Ubuntu <<>> example.com NS
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 44519
;; flags: qr rd ra; QUERY: 1, ANSWER: 2, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 4096
;; QUESTION SECTION:
;example.com.                IN      NS

;; ANSWER SECTION:
example.com.                 33095   IN      NS      a.iana-servers.net.
example.com.                 33095   IN      NS      b.iana-servers.net.

;; Query time: 4 msec
;; SERVER: 10.255.255.254#53(10.255.255.254) (UDP)
;; WHEN: Tue Oct 07 22:33:15 PDT 2025
;; MSG SIZE rcvd: 88
```

```

iu13@Nathan-Laptop:/mnt/c/Users/Siu13$ dig example.com SOA
; <<> DiG 9.18.39-0ubuntu0.24.04.1-Ubuntu <<> example.com SOA
; global options: +cmd
; Got answer:
; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 46950
; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1
; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 4096
; QUESTION SECTION:
;example.com.                IN      SOA
;
; ANSWER SECTION:
example.com.                3600    IN      SOA      ns.icann.org. noc.dns.icann.org. 2025082258 7200 3600 1209600 3600
;
; Query time: 50 msec
; SERVER: 10.255.255.254#53(10.255.255.254) (UDP)
; WHEN: Tue Oct 07 22:34:27 PDT 2025
; MSG SIZE  rcvd: 96

```

4.4)

```

C:\Users\Siu13>nslookup -type=CNAME www.wikipedia.org
Server:  node-1w7jr9n24twqzs2cg5ed4tjl3.ipv6.telus.net
Address:  2001:568:ff09:10c::67

Non-authoritative answer:
www.wikipedia.org      canonical name = dyna.wikimedia.org

```

```
C:\Users\Siu13>nslookup -type=TXT google.com
Server:  node-1w7jr9n24twqzs2cg5ed4tjl3.ipv6.telus.net
Address:  2001:568:ff09:10c::67

Non-authoritative answer:
google.com      text =

        "facebook-domain-verification=22rm551cu4k0ab0bxsw536tlds4h95"
google.com      text =

        "onetrust-domain-verification=de01ed21f2fa4d8781cbc3ffb89cf4ef"
google.com      text =

        "cisco-ci-domain-verification=47c38bc8c4b74b7233e9053220c1bbe76bcc1cd33c7acf7acd36cd6a5332004b"
google.com      text =

        "google-site-verification=wD8N7i1JTNTkezJ49swvWW48f8_9xveREV4oB-0Hf5o"
google.com      text =

        "apple-domain-verification=30afIBcvSuDV2PLX"
google.com      text =

        "v=spf1 include:_spf.google.com ~all"
google.com      text =

        "docusign=05958488-4752-4ef2-95eb-aa7ba8a3bd0e"
google.com      text =

        "docusign=1b0a6754-49b1-4db5-8540-d2c12664b289"
google.com      text =

        "google-site-verification=4ibFUgB-wXLQ_S7vsXVomSTVamu0XBiVAzpR5IZ87D0"
google.com      text =

        "MS=E4A68B9AB2BB9670BCE15412F62916164C0B20BB"
google.com      text =

        "google-site-verification=TV9-DBe4R80X4v0M4U_bd_J9cp0JM0nikft0jAgjmsQ"
google.com      text =

        "globalsign-smime-dv=CDYX+XFHUw2wml6/Gb8+59BsH31KzUr6c1l2BPvqKX8="
```